Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
AI and Application Development and Jera

Task -3

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Date of Assignment: 5-8-2026	Date/Time of Submission: 5-8-2026

Q.1Find the regression equation.Predict the price of a 1500 sq.ft. house. Predict the price of a 2500 sq.ft. house.

Area (sq.ft.)	Price (Lakhs)
600	32
800	40
1000	49
1200	58
1400	68
1600	79

Code:-

```
import pandas as pd
from sklearn.linear_model import LinearRegression
# Create dataset
data = {
   "Area": [600, 800, 1000, 1200, 1400, 1600],
   "Price": [32, 40, 49, 58, 68, 79]
}
df = pd.DataFrame(data)
# Independent and Dependent variables
X = df[["Area"]]
y = df["Price"]
# Train model
```

```
model = LinearRegression()

model.fit(X, y)

# Display Regression Equation

print("Regression Equation:")

print("Price =", round(model.coef_[0], 4), "* Area +", round(model.intercept_, 4))

# Predict prices

price1500 = model.predict([[1500]])

price2500 = model.predict([[2500]])

print("\nPredicted Price for 1500 sq.ft:", round(price1500[0],2), "Lakhs")

print("Predicted Price for 2500 sq.ft:", round(price2500[0],2), "Lakhs")

print("Afsha 25")
```

Output:-

```
...  Regression Equation:
     Price = 0.0469 * Area + 2.7905

     Predicted Price for 1500 sq.ft: 73.08 Lakhs
     Predicted Price for 2500 sq.ft: 119.93 Lakhs
     Radhika 17
```

Q.2 A company has collected the following employee data:

• Employee ID

• Employee Age

• Years of Experience

• Salary

You need to develop a Linear Regression model to predict an employee's salary.

Tasks

1. Identify the independent variable (X) and the dependent variable (Y).

2. Explain why you selected these variables.

3. Create a Pandas DataFrame using the given data.

4. Split the dataset into training and testing sets.

5. Train a Linear Regression model.

6. Predict the salary for an employee having 7 years of experience.

7. Display:

o Slope (Coefficient)

o **Intercept**

o **R2 Score**

8. Plot the regression line.

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
# Employee Dataset
data = {
    "EmployeeID":[101,102,103,104,105,106,107,108,109,110],
    "Age":[22,25,28,30,35,40,32,29,45,38],
    "Experience":[1,2,3,5,8,10,6,4,15,12],
    "Salary":[25000,30000,35000,45000,60000,70000,52000,40000,90000,80000]
}
df = pd.DataFrame(data)
print(df)
# Independent and Dependent Variable
X = df[["Experience"]]
y = df["Salary"]
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)
# Train Model
model = LinearRegression()
model.fit(X_train, y_train)
# Predict Salary
prediction = model.predict([[7]])
print("\nPredicted Salary for 7 years Experience =", round(prediction[0],2))
```

```
# Display Details
print("\nSlope (Coefficient):", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2 Score:", model.score(X_test,y_test))
# Plot Graph
plt.scatter(X,y,color="blue")
plt.plot(X,model.predict(X),color="red")
plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("Employee Salary Prediction")
plt.show()
```

Output:-

```
...    EmployeeID  Age  Experience  Salary
    0         101   22           1   25000
    1         102   25           2   30000
    2         103   28           3   35000
    3         104   30           5   45000
    4         105   35           8   60000
    5         106   40          10   70000
    6         107   32           6   52000
    7         108   29           4   40000
    8         109   45          15   90000
    9         110   38          12   80000

    Predicted Salary for 7 years Experience = 54117.65

    Slope (Coefficient): 4705.882352941176
    Intercept: 21176.470588235297
    R2 Score: 0.9880679800116606
```